DATA ITEM DESCRIPTION

Title: Acquisition and Sustainment Data Package (ASDP) Requirements for Production Part Approval Process (PPAP)

Number: DI-QCIC-82525
AMSC Number: 10644
DTIC Applicable: No
Preparing Activity: 11 (AFLCMC/EZSM)
Applicable Forms: None

Approval Date: 20260616
Limitation: N/A
GIDEP Applicable: No
Project Number: QCIC-2025-003

Use/Relationship: Defines the aviation and defense process requirements for the Acquisition and Sustainment Data Package (ASDP) Requirements Production Part Approval Process (PPAP). PPAP confirms that the production process has demonstrated the ability to produce products that consistently fulfill customer requirements.

This Data Item Description contains the content and intended use of information for the data deliverable resulting from the work tasks described in the solicitation.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

ASME Y14.41, Digital Product Definition Data Practices

Copies of ASME documents are available online at https://www.asme.org/.

ISO 9000, Quality Management Systems
SAE-AS6500, Manufacturing Management Program

Copies of ISO documents are available online at https://www.iso.org/.

SAE-AS9100, Quality Management Systems – Requirements for Aviation, Space, and Defense Organizations
SAE-AS9102, Aerospace First Article Inspection Requirement
SAE-AS9103, Variation Management of Key Characteristics
SAE-AS9110, Quality Management Systems – Requirements for Aviation Maintenance Organizations
SAE-AS9145, Requirements for Advanced Product Quality Planning and Production Part Approval Process
SAE-AS13004, Process Failure Mode and Effects Analysis (PFMEA) and Control Plans
SAE J1739, Potential Failure Mode and Effects Analysis (FMEA) Including Design FMEA, Supplemental FMEA-MSR, and Process FMEA

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Copies of SAE documents are available online at https://www.sae.org/.

DI-SESS-80776, Technical Data Package (TDP)

DI-SESS-81000, Product Design Data and Associated Lists

DI-SESS-82351, Acquisition and Sustainment Data Package (ASDP) Engineering Design Data and Associated Lists (EDDAL)

Copies of documents are available online at https://quicksearch.dla.mil/.

2. Format. The contractor's Production Part Approval Process (PPAP) data package must adhere to AS9145 Appendix D format, utilizing PDF-based forms for submission. Required data shall be mapped to model-based requirements and formatted as specified in the contract.

When adopting a model-based format, leverage the Department of Defense (DoD) and industry standard architecture frameworks such as the DoD Architecture Framework (DoDAF) or the Unified Profile for DoDAF/MoDAF (UPDM). Employing Systems Modeling Language (SysML) or the Unified Architecture Framework (UAF) is recommended to maximize information transferability, understandability, and reuse.

3. Content. The PPAP addresses the content specified in AS9145, Section 5, Appendix C, and Appendix D. Criteria and Criteria Explanations are listed below to help clarify the submission requirements. Data submissions include the following:

Note: Instructions to complete Production Part Approval Process (PPAP) for approval are described in AS9145.

Additional Factors:

a. *Additional information for content can be found in AS13004, Process Failure Mode and Effects Analysis (PFMEA).*

b. *Items determined not to be applicable (N/A) shall be explicitly identified as such in the contractor's submission.*

c. *To align PPAP requirements with the specific article's scope and complexity, utilize DD Form 1423, Contract Data Requirements List, for tailoring purposes.*

d. *If supplier PPAP submissions are required, this requirement shall be explicitly stated in the solicitation.*

The PPAP Contract Data Requirements List (CDRL) may contain items that duplicate other CDRLs specified in other sections of the SOW. When this occurs, the contractor shall indicate compliance by referencing the corresponding CDRL submissions.

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Production Part Approval Process File Contents			
Req.	**Production Part Approval Process (PPAP) Element**	**Product Development Process[1]**	**AS9145 Section References[2]**
1	**Design Records**	**2**	**3.9**
2	**Design Risk Analysis (e.g. Design Failure Mode and Effects Analysis (DFMEA); only applicable to design organization**	**2**	**3.10; 4.4.2; 4.4.4**
Explanation: Design Records define the product (system, part, component, subassembly, or assembly), including physical or electronic/digital drawings, electronic/digital models, software, or other associated information. Design risk analysis shall include design records addressing Key Characteristics (KCs), Critical Characteristics (CCs), Configuration Items (Cis), findings from DFMA, tolerance, stack-up analysis, or customer identified risks.			
3	**Process Flow Diagram**	**3**	**4.5.2; 4.5.3**
Explanation: This can include process flow diagrams described in Section 4.5.3. *Note: The Manufacturing Plan may already contain this information. (See SOW CDRL requirements).*			
4	**Process Failure Mode and Effects Analysis (PFMEA)**	**3**	**3.11; 4.5.2; 4.5.4**
Explanation: Process Failure Mode and Effects Analysis (PFMEA) is a methodology for assessing the weaknesses of production processes and the potential effects of process failures on the product being produced. The PFMEA shall be included in the Program's Quality Plan (See SOW CDRL requirements). *Ref. AS13004, Process Failure Mode and Effects Analysis (PFMEA) and Control Plans for details.*			
5	**Quality Control Plan**	**3**	**3.3; 4.5.2; 4.5.6; 4.5.7; 4.6.2; Appendix C**

[1] APQP has five phases starting with conceptual product needs and extending throughout the product life cycle. The actual duration of each phase will differ depending upon the scope and timing of the specific product and production development.

[2] Design Records - Per 5.2, "Production Part Approval Process File and Submission": Elements listed in Table 2 shall be included in the PPAP file, as applicable. There may be overlap between this data and other contract CDRL submissions.

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Explanation: The control plan details how product quality is controlled and confirmed at each stage of the manufacturing process, including the actions to be taken if the process becomes unstable or if nonconforming product is identified. (see AS9145, Appendix C).

Note: *The Program Quality Plan may already contain this information,* (*Ref. ISO9000, AS9100, OR AS9110 & SOW, CDRL requirements).*

6	**Measurement System Analysis (MSA)**	**4**	**3.14; 4.6.2; 4.6.3**

Explanation: Measurement Systems Analysis (MSA) is a tool used to analyze the variation present for each type of inspection, measurement, or test equipment. It is used to assess the quality of the measurement system and ensure that the variation in the measurement is minimal compared to the variation in the process.
Ref. AS9103, Variation Management of Key Characteristics.

7	**Initial Process Capability Studies**	**4**	**3.20; 4.6.2; 4.6.5**

Explanation: Collect and validate data and demonstrate that intended manufacturing processes and products conform to specified requirements. Methods shall include Process Capability Studies, comparing the output of an in-control process to the specification limits, expressed in terms of process capability (e.g., Cpk) or a process performance index (e.g., Ppk).

Note: This method is frequently used to ensure Key/Critical Characteristics meet design requirements (Ref. AS9100, AS9103 & AS6500 for additional information).

8	**Packaging, Preservation, and Labelling Approvals**	**4**	**4.6.6**

Explanation: Packaging and Preservation can include identification, handling, protection, contamination control, packaging, special handling, transportation, and handling of sensitive/hazardous products.

Note: Any special requirements shall be identified in the Program Quality Plan (Ref. AS9100 and SOW CDRLs for additional information).

9	**First Article Inspection Report (FAI)**	**4**	**4.5.2; 4.5.8**

Explanation:

a. Data associated with a comprehensive review of the initial production run of a new or revised part or assembly verifying that the manufacturing processes, from raw materials to finished products, that can meet engineering, design, quality and contract requirements (may incl. the Prime or Suppliers).

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b. When the customer imposes First Article Inspection (FAI) requirements beyond those defined in the applicable standard, the contractor shall ensure that evidence of compliance is generated, retained, and incorporated into the PPAP file in accordance with contractual provisions.

Ref. The Program Quality Plan (see AS9100, AS9102 and SOW CDRLs for additional information)

10	**Customer PPAP Requirements**	**4**	**4.6.3; 4.6.4**

Explanation: Complete the Production Part Approval Form(s), include a file containing objective evidence in support of PPAP requirements. Identify if fields are (R) Required or (CR) Conditionally Required.

Ref. AS9145 Appendix D for additional information.

Note: When the customer specifies additional PPAP requirements, such as Customer Demand Rate, Capacity Analysis, or other requirements, the organization shall include records of compliance for these requirements in the PPAP file.

11	**PPAP Approval Form (or equivalent)**	**4**	**4.6.7**

Explanation: Complete the Production Part Approval Form(s), including files containing objective evidence in support of PPAP requirements. Identify if fields are (R) Required or (CR) Conditionally Required.

Ref. AS9145 Appendix D for additional information.

End of DI-QCIC-82525.

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